

Exp-4 Develop a program for error detecting code using CRC-CCITT (16- bits)

```
import java.util.Scanner;

class crc
{
public static void main(String args[])
{
Scanner sc = new Scanner(System.in);

//Input Data Stream i.e creates object that reads data given by
keyboard

System.out.print("Enter data stream: ");

String datastream = sc.nextLine();

System.out.print("Enter generator: ");

String generator = sc.nextLine();

int data[] = new int[datastream.length() + generator.length()-1];

int divisor[] = new int[generator.length()];

//if not digit change it to digit

for(int i=0;i<datastream.length();i++)

data[i] = Integer.parseInt(datastream.charAt(i)+"");

for(int i=0;i<generator.length();i++)

divisor[i] = Integer.parseInt(generator.charAt(i)+"");

//Calculation of CRC

for(int i=0;i<datastream.length();i++)

{

    if(data[i]==1)
```

```

for(int j=0;j<divisor.length;j++)

    data[i+j] ^= divisor[j];

}

//Display CRC

System.out.print("The CRC code is: ");

for(int i=0;i<datastream.length();i++)

data[i] = Integer.parseInt(datastream.charAt(i)+"");

for(int i=0;i<data.length;i++)

    System.out.print(data[i]);

System.out.println();

//Check for input CRC code

    System.out.print("Enter CRC code: ");

datastream = sc.nextLine();

System.out.print("Enter generator: ");

generator = sc.nextLine();

data = new int[datastream.length() + generator.length()-1];

    divisor = new int[generator.length()];

for(int i=0;i<datastream.length();i++)

data[i] = Integer.parseInt(datastream.charAt(i)+"");

for(int i=0;i<generator.length();i++)

divisor[i] = Integer.parseInt(generator.charAt(i)+"");

//Calculation of remainder

for(int i=0;i<datastream.length();i++)

{

    if(data[i]==1)

        for(int j=0;j<divisor.length;j++)

            data[i+j] ^= divisor[j];

}

```

//Display validity of data

```
boolean valid = true;
```

```
for(int i=0;i<data.length;i++) if(data[i]==1)
```

```
{
```

```
    valid = false;
```

```
    break;
```

```
}
```

```
if(valid==true)
```

```
    System.out.println("Data stream is valid");
```

```
else
```

```
    System.out.println("Data stream is invalid. CRC error occurred.");
```

```
}
```

```
}
```